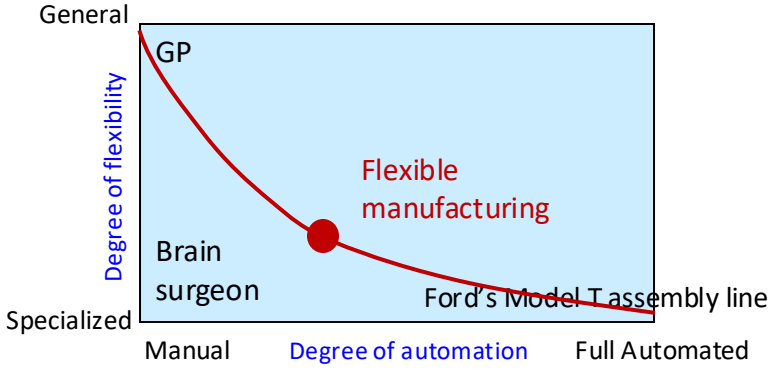
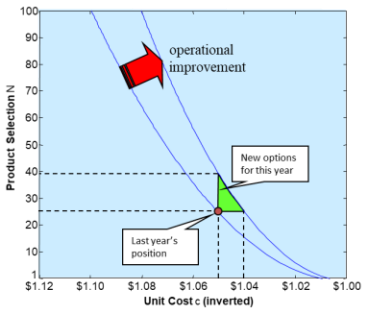
Types of Resources and Capacity

- Generalized vs. Specialized (or Dedicated) (degree of flexibility)
  - Benefits of specialization is efficiency and effectiveness (focus improves OE, Lily case)
  - Benefits of generalists and general-purpose equipment is endless flexibility
- Labor vs. capital (degree of automation)
- Flexible resources and capacity often fall somewhere in between; it's a spectrum and all about the trade-offs!



When do we need capacity flexibility?

- When flexibility offers alignment with your competitive positioning
  - Speed to market
  - Innovative products with short life cycles
  - Wide variety, high levels of customization



- When flexibility brings you closer to the efficient frontier
  - Same level of variety or innovation at lower cost
  - Higher level of variety or innovation at same cost
- When flexibility is less expensive than excess capacity
  - Use excess capacity if capacity is cheap, otherwise use flexibility
  - But remember, you need sufficient capacity to take advantage of flexibility

When do we need capacity flexibility?

- When
  - demand uncertainties are high
  - the correlation between the various demands is small or negative
  - forecasts improve significantly with time and flexibility allows the postponement of production decisions (information updating)
  - margins are different (flexibility allocation allows allocating more capacity to the higher margin product)

How is capacity flexibility valuable?

- Economies of scale (aggregating volumes)
  - Even at higher investment costs and higher marginal production costs, a flexible facility can be less expensive than separate dedicated facilities (each with their own investment cost) for a set of low volume products
- Risk pooling (aggregating volatilities, $\sigma_{pool} \leq \sigma_1 + \sigma_2$)
  - Reduced safety capacity (if it was positive to start with)
  - Increased service levels (fewer expected shortfalls)
- Information updating
  - Flexibility allows the delay of key decisions and the collection of more information
    - e.g. the advantage of the "flexible → specialized" solution in Eli Lilly case with respect to the go/no-go decision and size decision for the specialized plant
- Flexibility allocation (contingent decision-making, more than risk pooling)
  - Increased revenues (fewer losses) and reduced variance in profits
    - e.g. if the flexible capacity is insufficient to meet all the demand, allocate it to the most profitable products first

→Capacity flexibility makes variety and innovation more profitable

Benefits of flexibility

- Flexibility is about the ability to adopt change
- At the plant level, flexibility is the ability to adapt or change and the objectives are to
  - Increase scope flexibility: breadth of activities
  - Increase agility: speed of changeovers, could include scale flexibility
  - Achieve robustness: performance is uniform over the range of activities performed
- Operations strategy tailors the appropriate type, amount, and configuration of flexibility for alignment
  - Scope is desirable for strategies that emphasize variety and customization
  - Agility is important when competing on speed and timeliness
  - Robustness is important for strategies that emphasize cost and quality

Challenges of flexibility

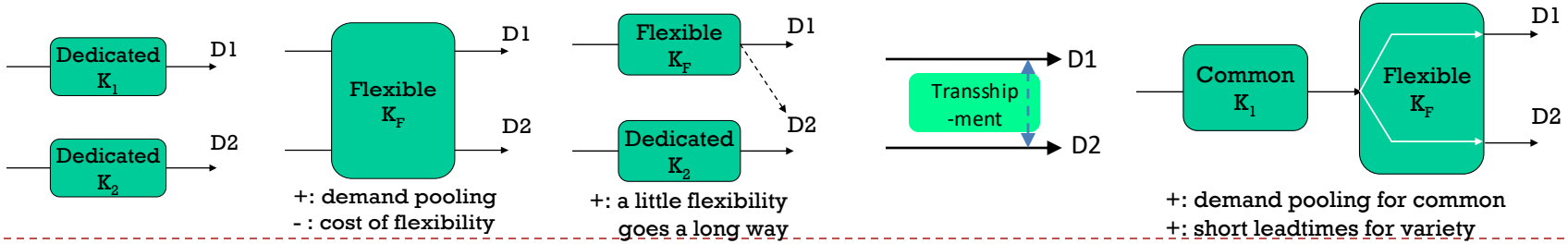
- Flexibility is expensive and difficult to value
- Flexibility is not well understood
  - It is often unclear which features of a process must be changed to enhance its flexibility, e.g., computer integration is often not correlated with increased scope or agility in SC, operators once can handle extreme demands
    - Flexibility is driven more by people and management than by equipment, e.g., plant flexibility can often be explained by managerial action rather than structural differences in processes
    - Scope flexibility typically decreases with scale, agility is not correlated with scale
- Flexibility is not used appropriately
  - FMS promises have gone unfulfilled when they are not well used (Jaikumar 1986)
- Customer perception may preclude using flexibility
  - Jaguar X-type is not a real Jaguar as it shared Ford's Mondeo platform
  - Lexus ES300 shared the Toyota Camry platform

Flexibility through *product design*

- Standard or universal design
- Component commonality and platform sharing
  - e.g ,one component can be used in 5 different models
  - This allows the production of variety (5 models) at lower cost, higher conformance quality, faster response time, etc.
- Modular design
  - e.g., instead of just one component, we now create five different versions of this component BUT all of them fit in all the 5 original models, thereby creating 25 varieties.
  - We can thus increase product variety without taking large hits in other competencies
- Adaptable design
  - Product can be changed easily by designer/manufacturer, dealer, or customer to fit a variety of customer demands, e.g. software

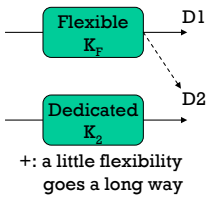
Flexibility through *process design*

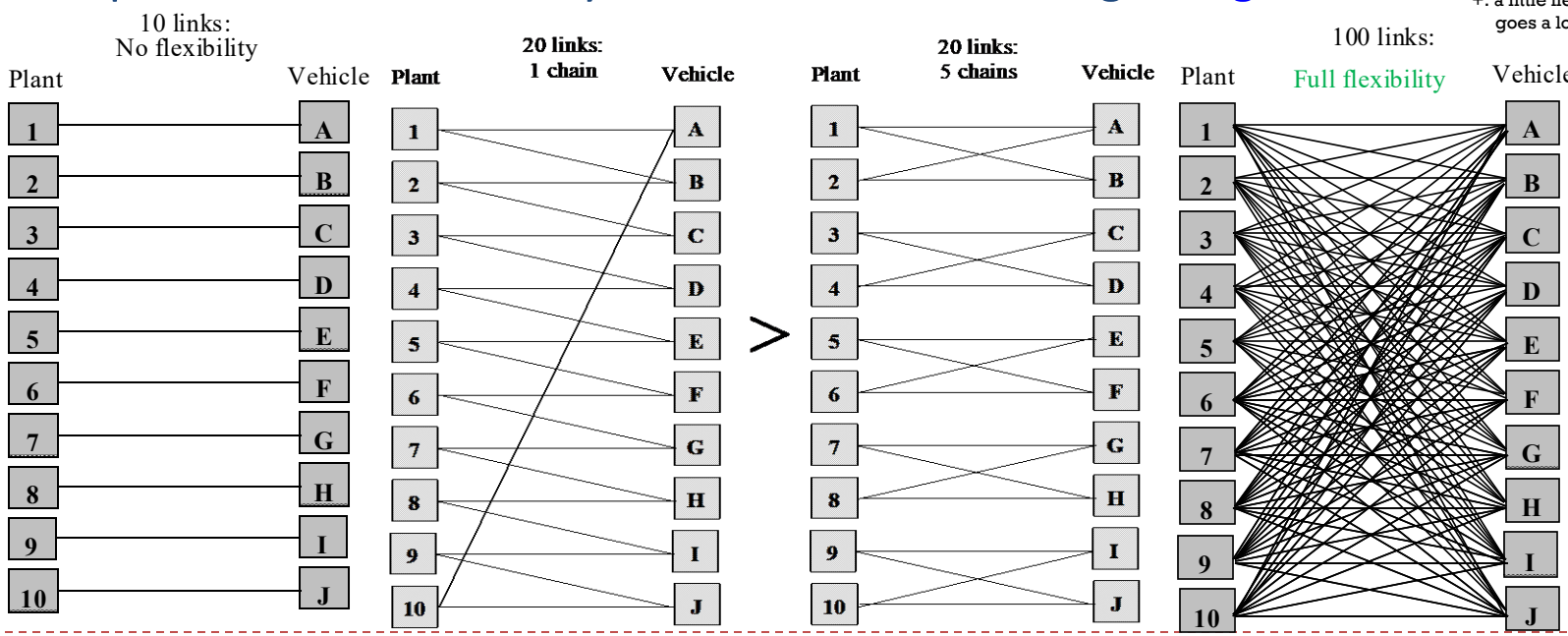
- Dedicated facilities and resources
- Facility and resource sharing
  - e.g., A common flexible capacity for two products – requires less safety capacity because of pooling
- Facility and resource substitution
  - e.g., One flexible asset and one dedicated asset. This mix of flexible and dedicated capacity can combine the pooling power of flexible capacity with the lower investment cost of dedicated capacity.
- Transshipment
  - Relocating assets
- Postponement (for delayed differentiation)
  - Delayed differentiation combines the advantages of product designs that have common components and differentiator modules with a process in which the differentiating elements are added at the end of the process in order to reduce the time between product specification and delivery



Tailored Process Flexibility

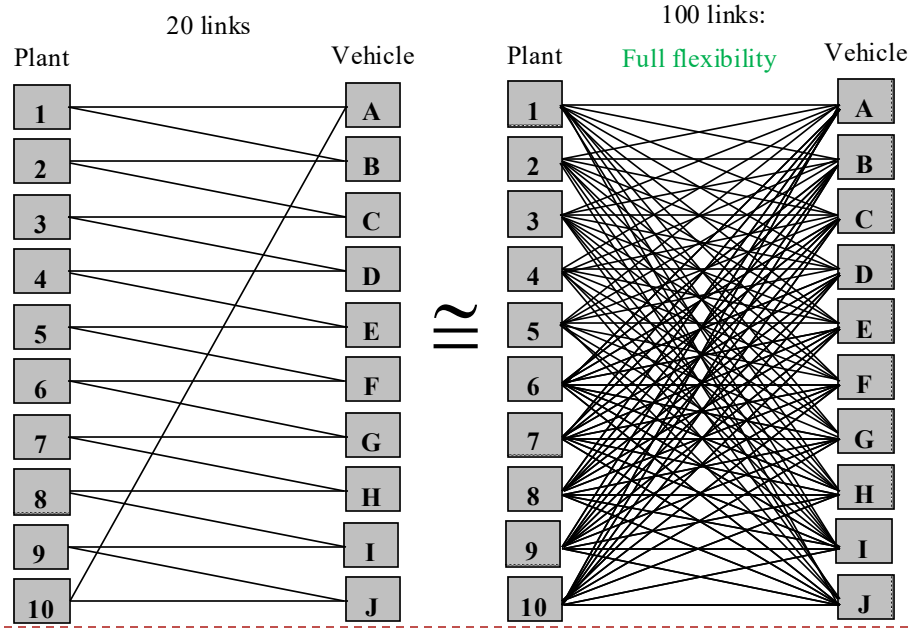
- Flexibility exhibits decreasing returns but increasing costs so that an intermediate amount of flexibility maximizes value
- Capacity flexibility can be achieved through resource chaining (sharing and substitution in process design)
  - A chain is a group of plants and products connected via links
  - Important Note: flexibility is more effective through long chains





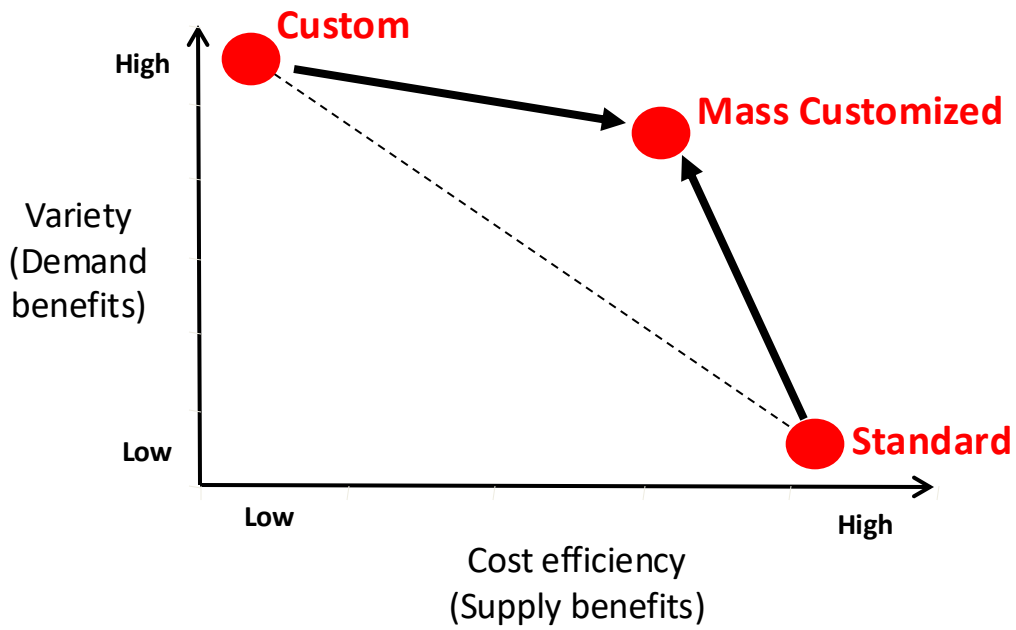
A little flexibility goes a long way

- A configuration with 20 links can produce nearly the results of total flexibility if it constructs correctly in one long chain
  - Simulation results: with CV of at most 0.33, the ratio of the expected sales from the long chain system to the total flexibility system is 89% for Uniform and 96% for Normal demand distribution
- The value of flexibility exhibits decreasing returns, “a little flexibility goes a long way” and is often preferred to full flexibility.



Mass Customization (MC)

- Mass customization tailors a product or service to *each* customer's wishes within certain limits, and producing and delivering it with a flexible flow process with negligible changeovers
  - It's not mass production that customers are looking to escape; it's mass products.



Technology behind Mass customization

- Product technology
  - Uses modularity and platform sharing along with adaptable design
    - The “product” really is a menu of options: e.g., 10 components, each having 5 options
    - Rationalized standardization: smart choice of the options is key
- Process technology
  - Uses Flexibility through postponement
    - At least the last process step must be MTO (it really is ATO) and must be able to support a batch size of 1 (i.e. negligible change overs)
  - Key is
    - To postpone differentiation fan-out as long as possible
    - To have information capability to support customized last step
    - for customer to be willing to *wait* for the last step
- Coordination & Information technology
  - systems allow customers to self-design their product and transmit that information automatically to the process with billing and tracking
- Transportation technology
  - delivery or self pick

Guidelines for capacity flexibility

- How much flexibility?
  - Process design: a little flexibility goes a long way
- Reduce the cost of flexibility using
  - setup time reduction,
  - adaptable capacity,
  - postponement and
  - reactive capacity
- Integrate flexibility strategy with new product strategy
- Be clear on the role of flexibility
  - Highly volatile demands (changeovers)
  - Low volume (technology transfer skills)
  - Prototyping (PM skills)

Learning Objectives for today

- Be able to
  - Describe the benefits and challenges of different types of capacity and flexibility
  - Describe how flexibility can be achieved through product and process design and mass customization
  - Tailor and optimize the capacity portfolio of a multi-product firm in a spreadsheet
    - Eli Lily case study